Lecture 14: Synthesis
Solar System in Context, Astrobiology

Planetary Systems

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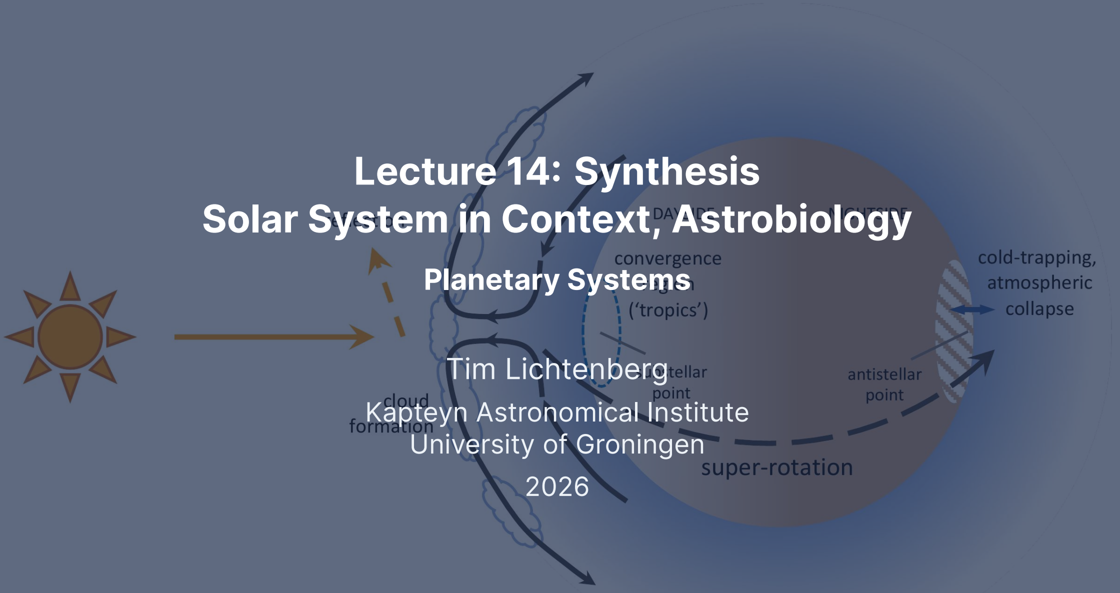


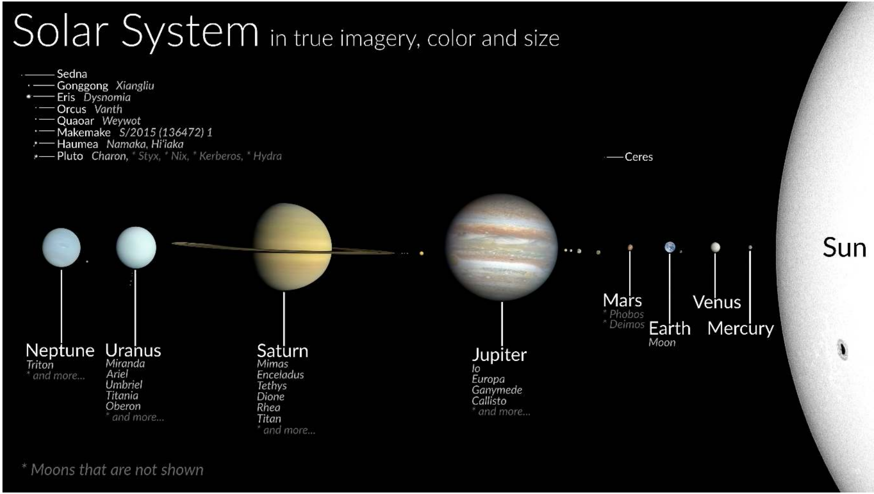
Tidally locked rocky planet atmospheric circulation. Wordsworth & Kreidberg (2022)

Learning Objectives

By the end of this lecture, you will be able to:

- Place our solar system within the observed exoplanet population
- Evaluate habitability as a coupled *systems* property, not a single condition
- Derive classical habitable-zone boundaries from stellar luminosity and liquid water
- Identify the most promising life-detection targets for the next decade

Habitability is a trajectory, not a snapshot.



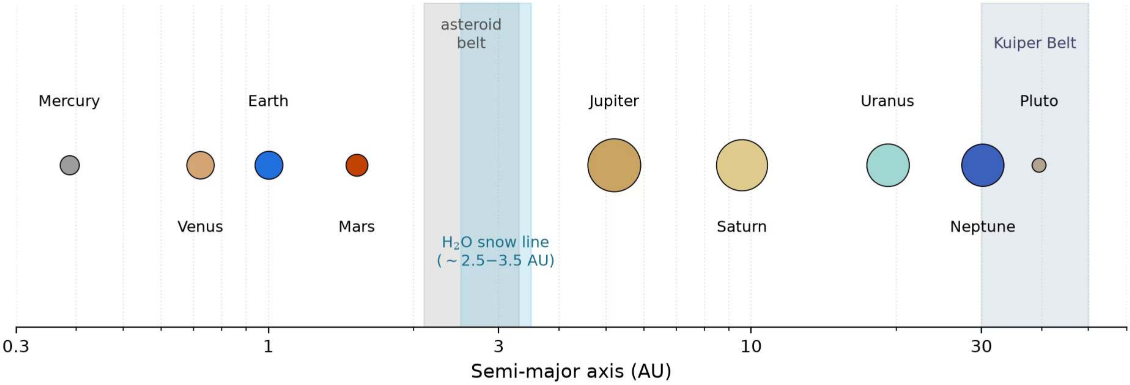
Composite of the principal bodies of the solar system grouped by region at true relative size, from terrestrial planets to gas and ice giants and Kuiper belt objects. Credit: CactiStaccingCrane (Wikimedia Commons, CC BY-SA 4.0); NASA/ESA/ISRO.

The Thread of the Course

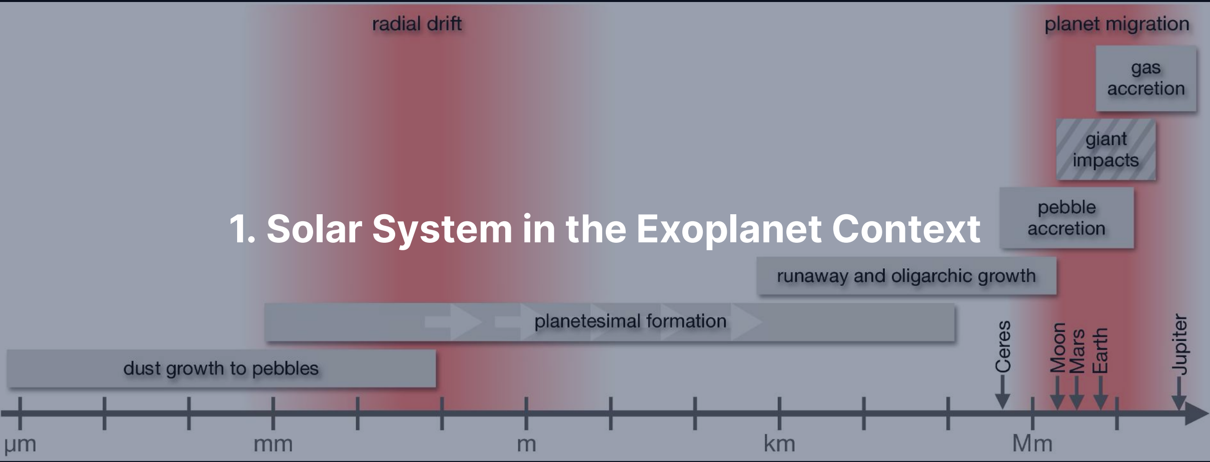
The same physical laws govern planetary formation, evolution, and climate throughout the galaxy.

- L1–L8: accretion, differentiation, internal heat engines, and atmosphere-surface coupling
- L9–L12: terrestrial and giant planet evolution traced body by body
- L13: over 6000 exoplanets revealing architectural diversity

Today: Step back to evaluate the solar system in context and test astrobiological habitability.



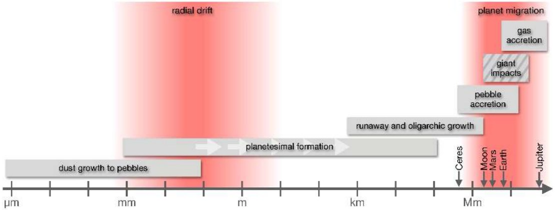
Architecture of the Solar System on a logarithmic semi-major axis scale with symbol area scaled to radius. Shaded spans mark the asteroid belt, Kuiper Belt, and water snow line region. Credit: Course-original figure; NASA NSSDC.



1. Solar System in the Exoplanet Context

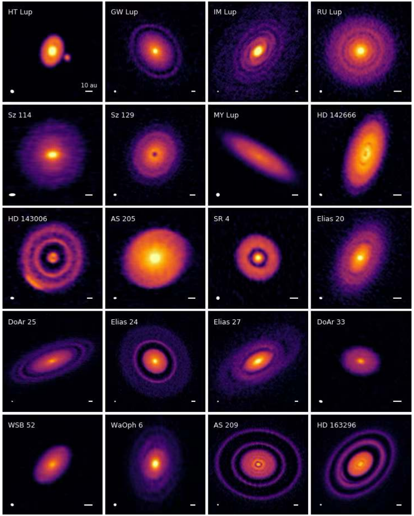
Dust-to-planet growth. Drążkowska et al. (2023)

Formation Theory Meets Observation

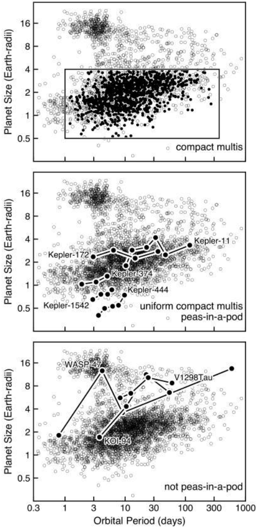


- Dust → pebbles → planetesimals (km-scale bodies) → embryos.
- Streaming instability (pebble clumping) + pebble accretion + runaway envelope accretion.
- Disk lifetime 3–5 Myr is the deadline.
- ALMA disk substructure is direct observational input.

ALMA's DSHARP Reveals Forming-Planet Signatures



- 240 GHz continuum, 20 nearby disks.
- Concentric rings, gaps, asymmetries: nearly ubiquitous.
- Most are interpreted as signatures of in-progress planet growth.
- Andrews+ 2018.

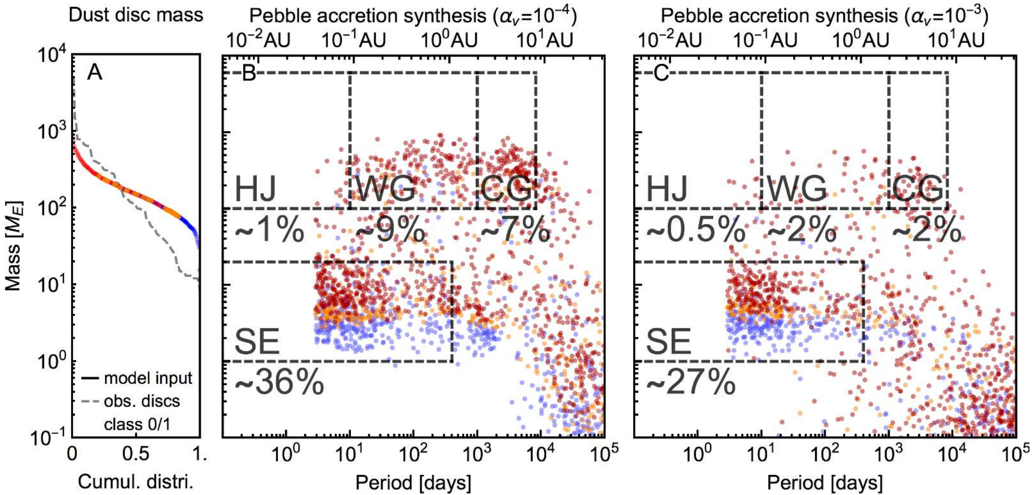


Transiting census; box and filled circles are compact multis (top), five uniform ones (middle), three non-uniform (bottom). Compact multi-planet systems exhibit striking architectural uniformity. Weiss et al. (2023, PPVII).

Where the Solar System is Atypical

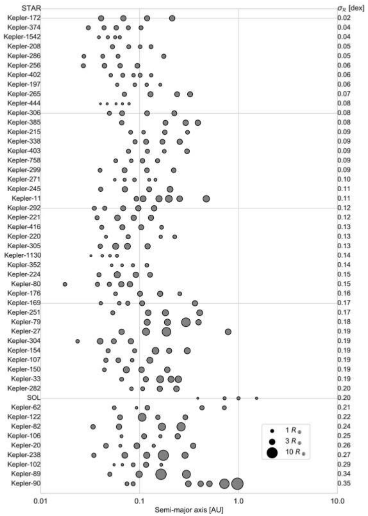
The architecture of the solar system diverges noticeably from typical planetary systems discovered by exoplanet surveys.

- No super-Earths or sub-Neptunes: the most common exoplanet size class is absent
- No close-in giant planets: hot Jupiters orbit ~ 0.5–1% of Sun-like stars, hot Neptunes are rarer still
- Outer giants on wide, circular orbits, while inner terrestrials lack compact uniformity



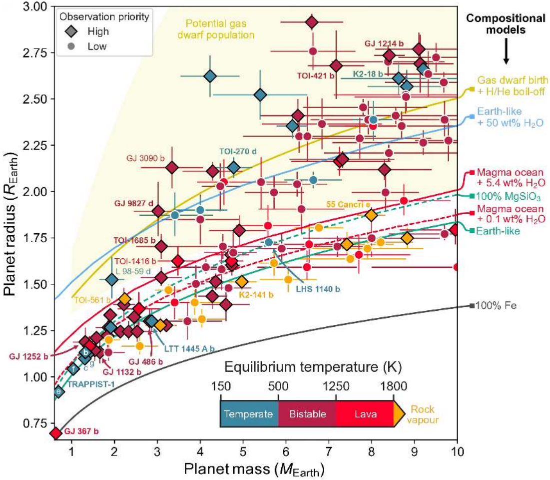
Population synthesis predictions for planet mass versus orbital period under planetesimal and pebble accretion. Both pathways readily form super-Earths and close-in giants that the solar system lacks. Credit: Drążkowska et al. (2023).

Peas-in-a-Pod: Solar System Doesn't Fit



- Compact multi-systems with ≥ 4 transiting planets within 1.52 AU.
- Ranked by uniformity of planet sizes.
- Solar system: *bottom quintile* of size uniformity.
- Inner solar system did not produce the typical compact-multi architecture.

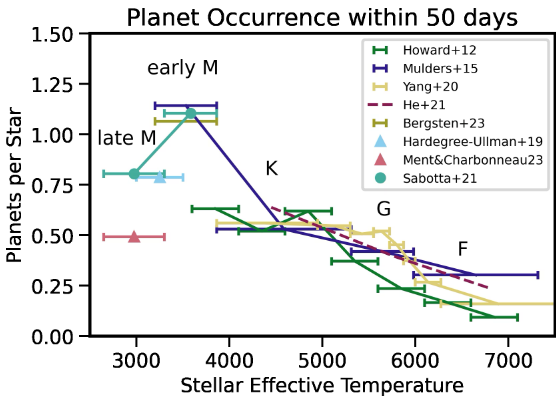
Mass-Radius: Where Each Planet Sits



- EOS tracks: 100% Fe, Earth-like, $MgSiO_3$, water-rich, magma + H_2O, gas dwarf.
- Solar system terrestrials sit on the Earth-like curve.
- Close-in super-Earths sit there too: *stripped cores* that lost gas envelopes.
- Larger planets carry volatile envelopes that swell radii.

Source: Lichtenberg et al. (2025)

Occurrence Climbs Toward M Dwarfs



- Small planets (1–4 $R_\oplus$, $P < 50$ d) per star vs T_{eff}.
- Occurrence roughly 2× higher around early M dwarfs than F dwarfs.
- Trend continues into late M dwarfs.
- Earth-sized planets are common around the most numerous stars.

Source: Mulders (2024)

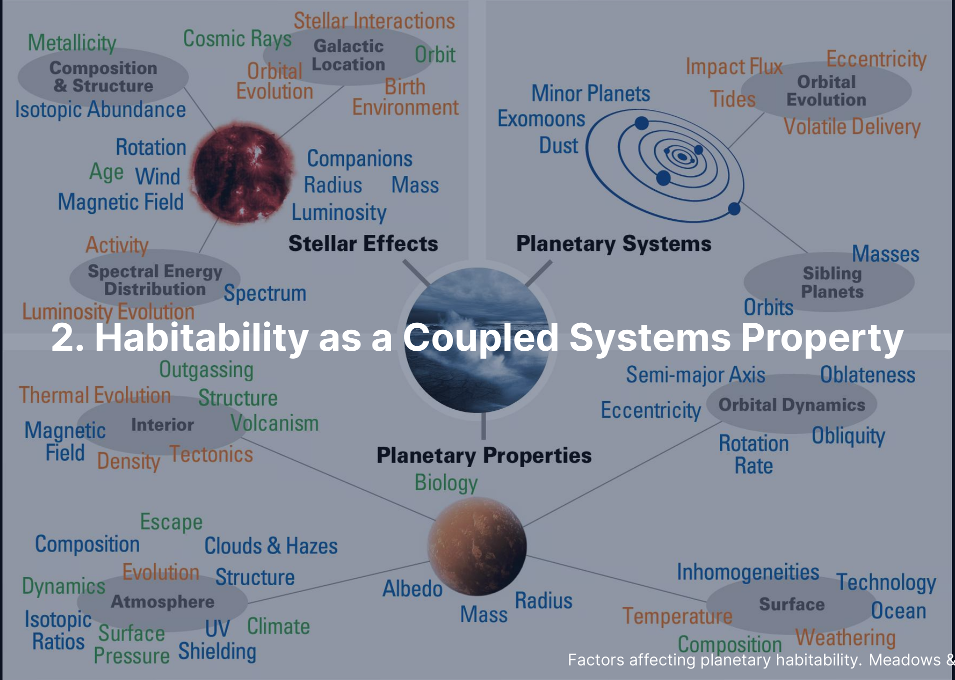
Is the Solar System Rare?

Credit: C. Letelier/ESO

Whether our architecture is rare remains open because 1 AU terrestrial analogues sit at current survey detection limits.

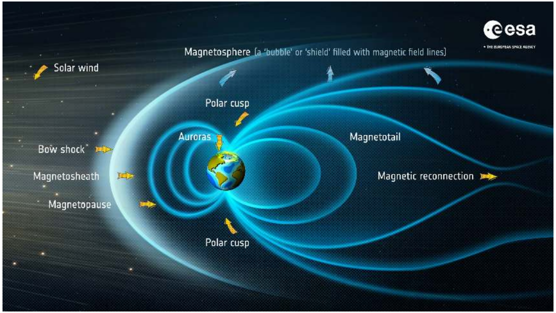
- Kepler four-year baseline limited sensitivity to long-period orbital architectures
- Inner solar system is structurally atypical compared to known compact multi systems
- PLATO (2027) and Gaia DR4 (late 2026) will resolve true occurrence

2. Habitability as a Coupled Systems Property



Factors affecting planetary habitability. Meadows & Barnes (2018)

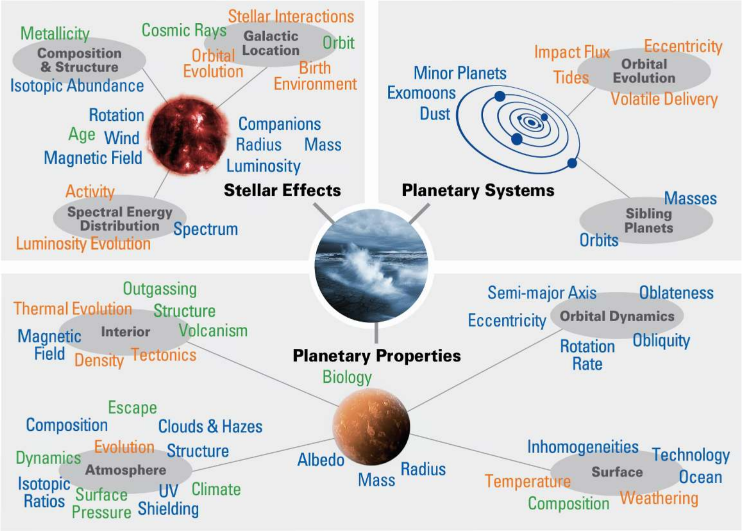
The Stack of Requirements



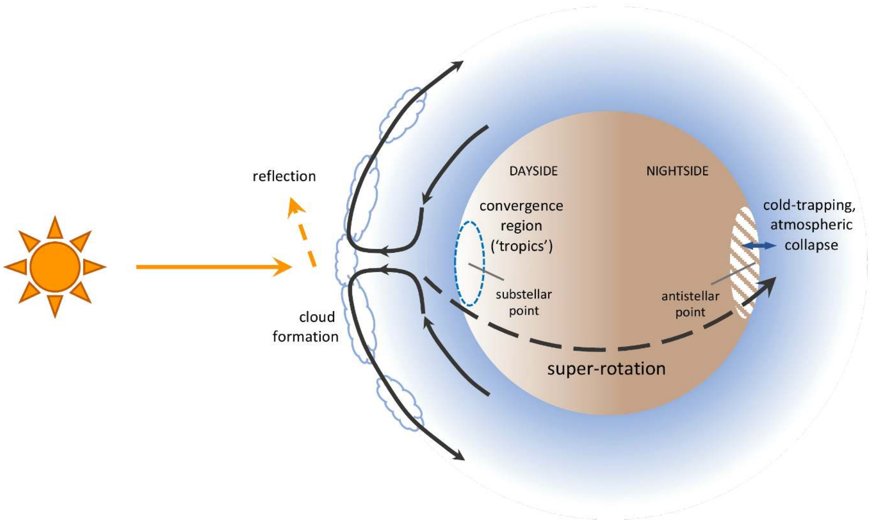
Credit: ESA, CC BY-SA 3.0 IGO

Habitability is not an isolated checklist but a coupled stack of astrophysical, geophysical, and geochemical conditions.

- Stellar host: stable luminosity over Gyr, benign stellar activity, and orbit in habitable zone
- Geophysical engine: bulk mass driving internal dynamo and tectonic volatile recycling
- Geochemical envelope: atmospheric pressure, escape rate, and surface liquid water stability
- Biosphere: feeds back on weathering and atmospheric redox

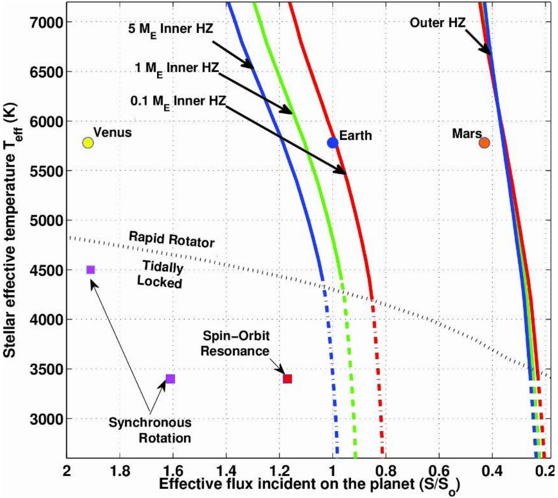


Factors affecting habitability: stellar, planetary, and orbital properties jointly shape the surface environment; breaking any coupling can terminate surface water stability. Blue: observable; green: model-interpreted; orange: theoretical. Meadows & Barnes (2018).



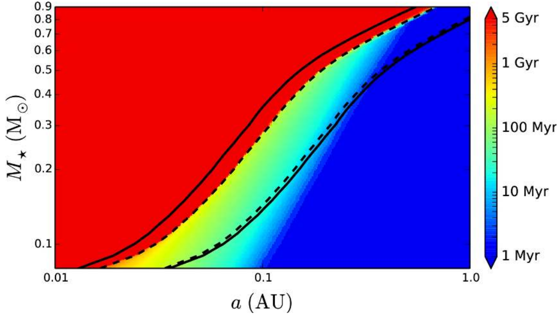
Atmospheric circulation on a tidally locked rocky planet: substellar convection and equatorial super-rotation transport heat, while nightside cold traps risk atmospheric collapse. Wordsworth & Kreidberg (2022).

The HZ is Not a Line



- Inner edge: runaway greenhouse limit.
- Outer edge: maximum $\mathrm{CO_2}$ greenhouse limit.
- Two planets at identical flux can have radically different climates.
- Time-dependent: stars brighten over main-sequence lifetime.
- Orbiting in the habitable zone today does not guarantee a habitable history.

M Dwarf HZ: Pre-MS Runaway Phase



- Duration of runaway greenhouse on a water-bearing planet vs $M_\star$.
- Late M dwarfs: up to a Gyr of runaway before reaching the main sequence.
- Planet starts *inside* runaway zone and loses initial water inventory.

Source: Luger & Barnes (2015)

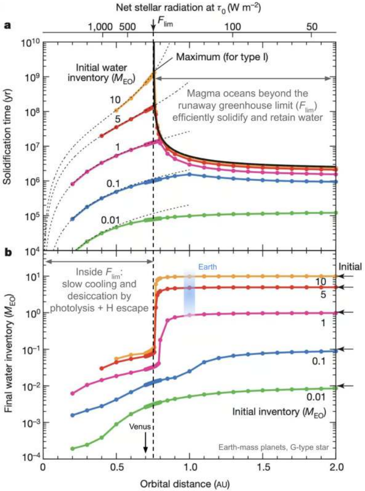
3. Magma Oceans, Water Delivery, Tectonic Thermostat

Magma-ocean differentiation. Lichtenberg et al. (2023)

Why ”Delivery” Isn’t the Right Frame

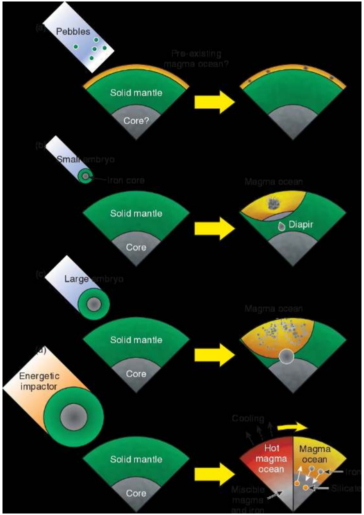
Terrestrial volatile inventories diverge because planetary differentiation and subsequent evolution matter far more than initial delivery.

- Earth, Venus, and Mars received similar initial accreted volatile budgets
- Magma ocean partitioning sets core, mantle, and steam atmosphere reservoirs
- Evolutionary controls: early hydrodynamic escape, mantle outgassing, and stellar brightening



Two types of terrestrial planet from magma ocean evolution. Inside critical flux, Type II planets lose water to hydrodynamic escape before solidifying; Type I planets further out solidify quickly and retain oceans. Credit: Hamano et al. (2013).

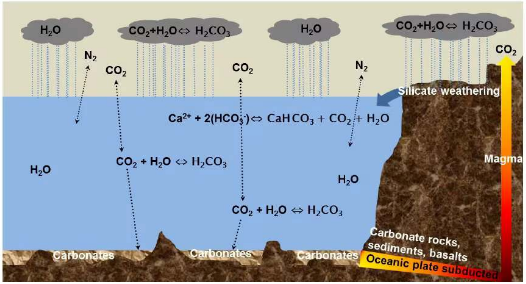
Magma Ocean Differentiation Sets Boundary Conditions



- Initially molten mantle separates from sinking metallic iron core.
- Volatiles partition between liquid silicate and steam atmosphere.
- Sequestered fraction depends on mantle redox, solidification time, and depth.
- Identical initial budgets yield radically different atmospheric masses.

Source: Lichtenberg et al. (2023)

Tectonic Regime: The Long-Term Thermostat

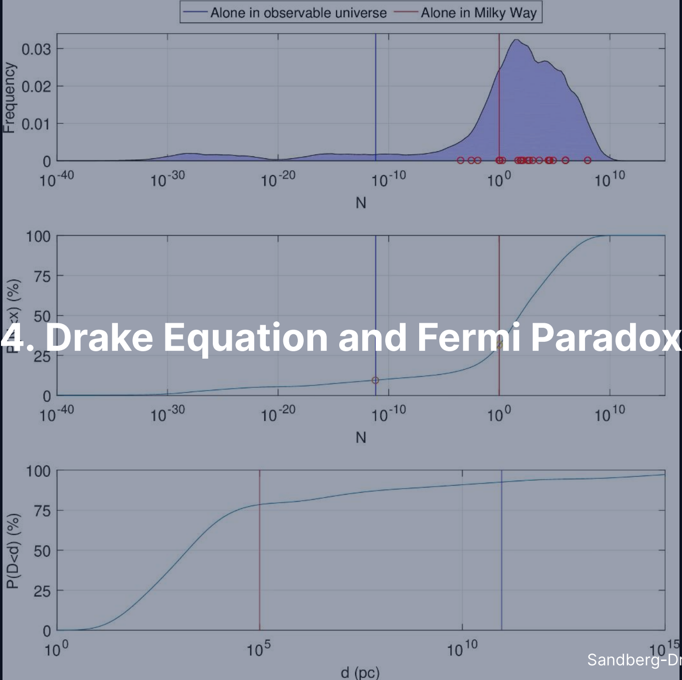


Credit: Lammer et al. (2018)

The carbonate-silicate cycle acts as a negative climate feedback that buffers surface temperature over geological timescales.

- Volcanic $\mathrm{CO_2}$ outgassing balanced by temperature-dependent silicate weathering
- Subduction recycles carbonate minerals into mantle; failure produces Venus-like 92 bar
- Requires surface liquid water, active volcanism, and tectonic mantle recycling

4. Drake Equation and Fermi Paradox



Sandberg-Drexler-Ord Drake posterior (2018)

The Drake Equation: A Framing Tool

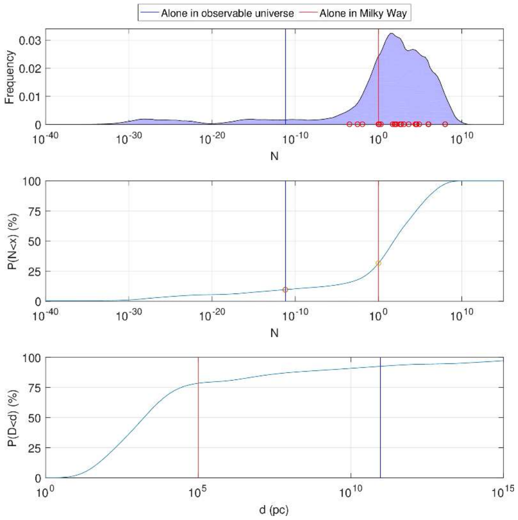
$$N = R_\star \cdot f_p \cdot n_e \cdot f_l \cdot f_i \cdot f_c \cdot L$$

$R_\star$ star-formation rate, f_p fraction with planets, n_e habitable planets per system, f_l life, f_i intelligence, f_c communication, L lifetime of a signalling civilisation.

- Astrophysical factors are now measured: $R_\star$, $f_p \sim 1$, $n_e \approx 0.1$–0.6 (Bryson+ 2021)
- Biological and social terms (f_l, f_i, f_c, L) span 4–10 orders of magnitude
- Factors co-evolve rather than acting as independent pipeline stages

The Drake equation is a tool for framing ignorance, not a numerical estimator.

Sandberg+ 2018: Honest Posterior Dissolves Fermi



- Monte Carlo sampling over full published parameter ranges.
- Resulting probability distribution for N is *bimodal*.
- Substantial probability mass sits below $N = 1$ in the Milky Way.
- Apparent cosmic silence is unsurprising once priors are honest.

Source: Sandberg, Drexler & Ord (2018)

Why Bimodal? Log-Sum of Broad Priors

Taking the logarithm turns the Drake product into a sum of broad priors, so $\log N$ is spread over many orders of magnitude.

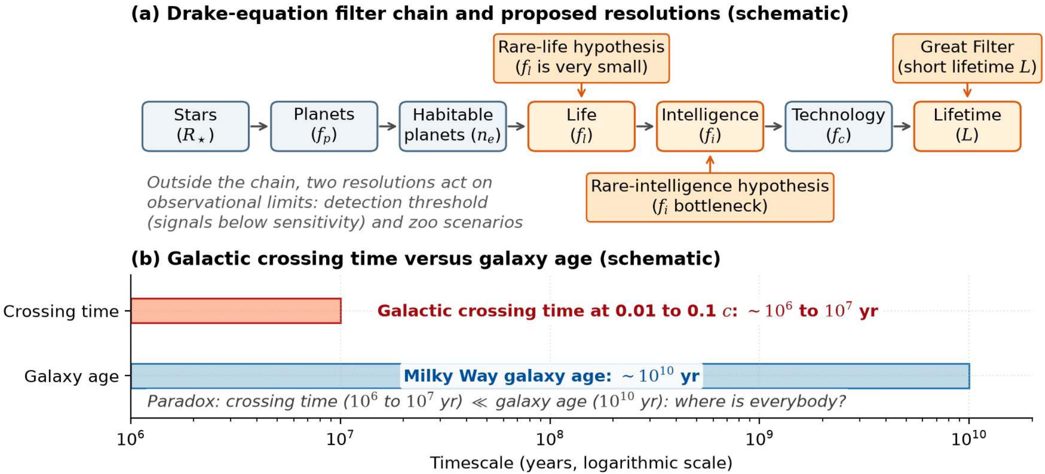
$$\log_{10} N = \log_{10}(R_\star f_p n_e) + \log_{10} f_l + \log_{10} f_i + \log_{10} f_c + \log_{10} L$$

- Unconstrained factors have log-uniform priors spanning many decades
- Log-sum of broad, skewed priors: $\log N$ spreads over tens of decades, and the long low tail of f_l makes the result bimodal
- Posterior width spans ~ 30 decades, placing high probability on $N < 1$

Fermi Paradox: A Question, Not an Answer

The apparent silence of the cosmos acts as an intuition check on subjective priors rather than physical evidence.

- Rare origin: abiogenesis ($f_l \ll 1$) or intelligence ($f_i \ll 1$) is exceptionally improbable
- Great filter: civilisations self-terminate rapidly, making lifetime L short
- Observational threshold: transmissions exist but lie below current detection limits



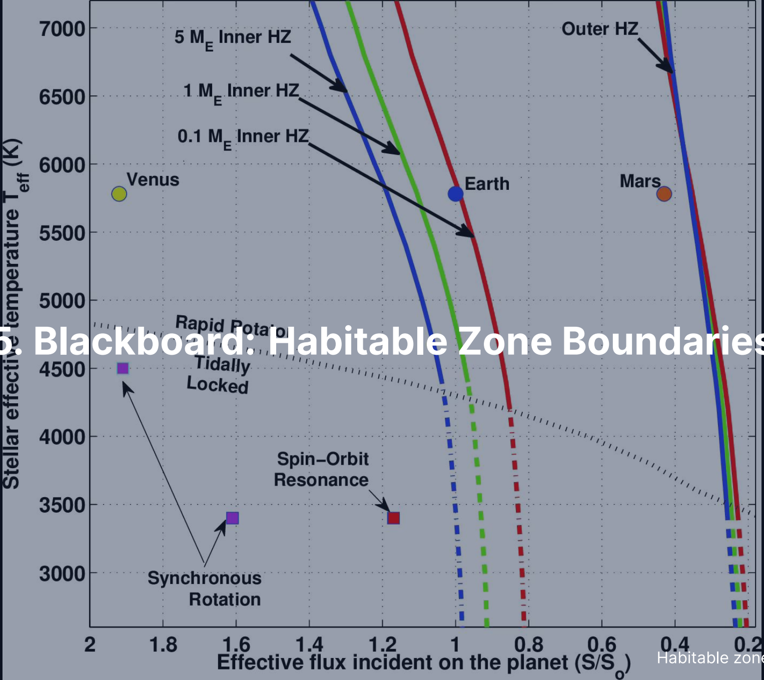
The Fermi paradox as a chain of filters: rare life, rare intelligence and the Great Filter act on single Drake factors; a galactic crossing takes 10^6 to 10^7 yr against a 10^{10} yr galaxy age. Course figure.

Break

End of Part 1: Solar System Context and Habitability
Part 2: Habitable Zone Derivation, Astrobiology, Life Detection

Earthrise, Apollo 8 (1968). Credit: NASA

5. Blackboard: Habitable Zone Boundaries



Habitable zone vs T_{eff}. Kopparapu (2014)

Setup: Stellar Flux and Equilibrium Temperature

Star with bolometric luminosity $L_\star$, planet at distance d:

$$F_\star(d) = \frac{L_\star}{4\pi d^2}$$

Bond albedo A_B (reflected fraction), rapidly rotating sphere, equilibrium:

$$T_{\mathrm{eq}}(d) = \left[\frac{L_\star(1 - A_B)}{16\pi\sigma\epsilon\, d^2}\right]^{1/4}$$

For Earth: $A_B = 0.3$, $\epsilon = 1$, $L_\star = L_\odot$, $d = 1$ AU $\to T_{\mathrm{eq}} \approx 255$ K.

Surface is 288 K from ~ 33 K greenhouse warming. *To the board.*

Result: Habitable Zone Boundaries

Effective flux $S_{\mathrm{eff}} = F_\star(d)/S_\oplus$ and the boundary distances:

$$d = 1\ \mathrm{AU}\sqrt{\frac{L_\star/L_\odot}{S_{\mathrm{eff}}}}$$

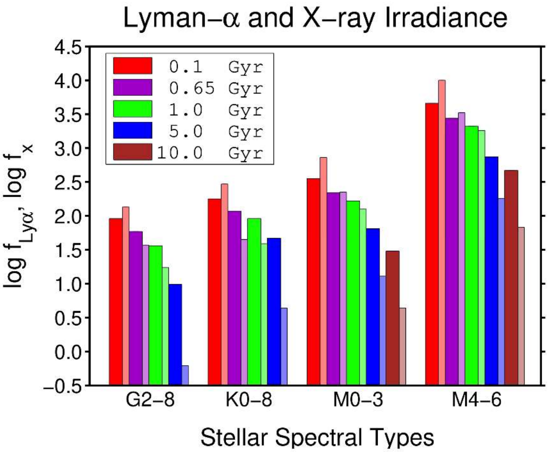
- Inner edge: runaway greenhouse (Simpson-Nakajima, Lecture 9), $S_{\mathrm{in}} \approx 1.06$, ≈ 0.97 AU for the Sun
- Outer edge: maximum $\mathrm{CO_2}$ greenhouse, $S_{\mathrm{out}} \approx 0.35$, ~ 1.69 AU for the Sun
- Main-sequence lifetime $t_{\mathrm{MS}} \approx 10\ \mathrm{Gyr}\,(M_\star/M_\odot)^{-2.5}$ sets how long the zone lasts

HZ Across Stellar Types

Habitable zone distance scales with stellar luminosity: $d \propto \sqrt{L_\star}$.

- **G dwarf** ($L_\odot$): d = 0.97–1.69 AU; main-sequence lifetime ~ 10 Gyr
- **K dwarf** ($0.3\,L_\odot$): d = 0.53–0.93 AU; lifetime 24 Gyr (orange-dwarf sweet spot)
- **M dwarf** ($5 \times 10^{-4}\,L_\odot$): d = 0.022–0.038 AU; lifetime 10^{12} yr, but tidally locked with long pre-MS runaway

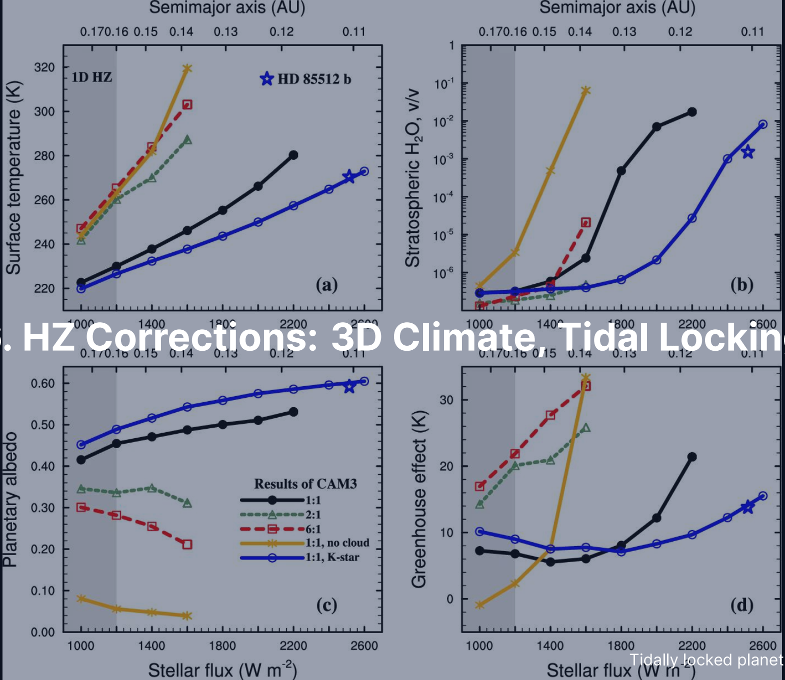
XUV at the HZ: M Dwarfs are Hostile



- Lyman-α and X-ray fluxes at Earth-equivalent distance.
- Flux is > 100–500× higher around M dwarfs than G dwarfs.
- X-rays drop 500× over 5 Gyr; Lyman-α drops only 5–10×.
- Early K dwarfs offer long lifetimes without M-dwarf XUV extremes.

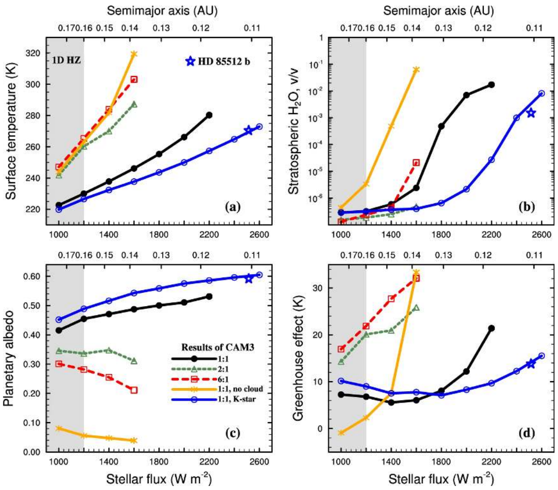
Source: Cuntz & Guinan (2016)

6. HZ Corrections: 3D Climate, Tidal Locking



Tidally locked planet clouds. Yang et al. (2013)

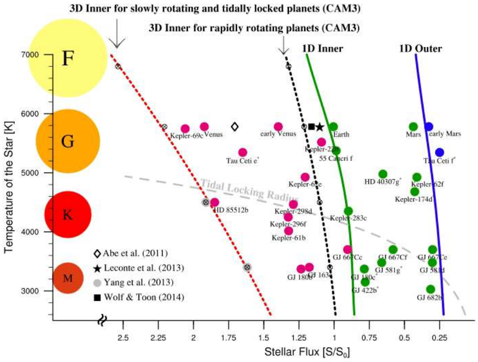
3D GCMs Shift Inner Edge of M Dwarf HZ



- 3D global climate model of tidally locked planet with clouds.
- Persistent substellar convection lifts thick convective water clouds.
- The planetary albedo rises to ~ 0.6 and reflects more stellar flux.
- Inner edge tolerates ~ 2× higher flux than 1D models predict.

Source: Yang et al. (2013)

Where M Dwarf HZ Planets Actually Sit



- Shields+ 2016; stellar flux increases toward the left.
- Solid curves: 1D Kopparapu inner and outer boundaries.
- Red dotted: 3D inner edge tolerates 1.7–2.2× 1D flux.
- Grey dashed: tidal-locking radius encloses almost all M-dwarf HZ planets.

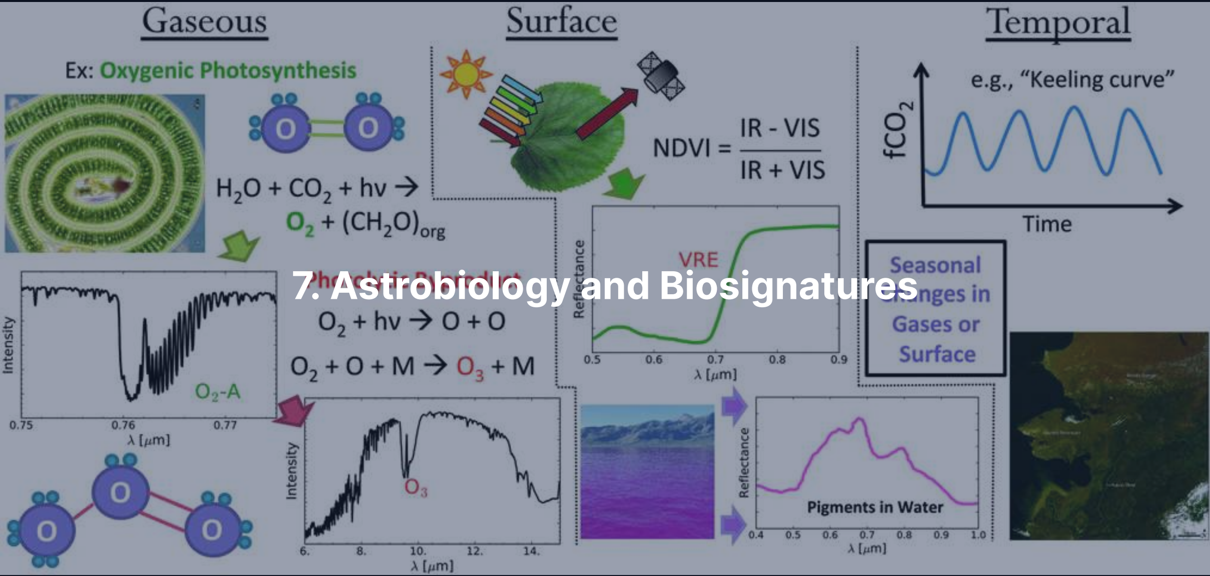
Tidal-Locking Timescale

Tidal despinning scales steeply as a^6, ensuring that close-in habitable planets around low-mass stars synchronize their rotation rapidly.

$$\tau_{\mathrm{lock}} \sim \frac{\omega_0\, a^6\, I_p\, Q_p}{3\, G\, M_\star^2\, k_{2,p}\, R_p^5}$$

- Earth at 1 AU has $\tau_{\mathrm{lock}} \sim 10^{12}$ yr, far exceeding universe age
- TRAPPIST-1 b ($a \approx 0.011$ AU) synchronises in $\lesssim 10^7$ yr
- With system age ~ 8 Gyr, all known M-dwarf HZ worlds are locked

7. Astrobiology and Biosignatures



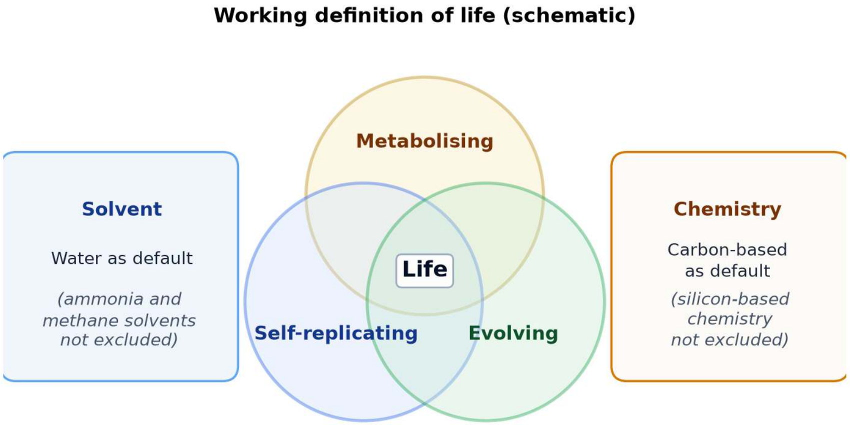
Biosignature classes. Schwieterman et al. (2018)

Defining Life and Its Origin Hypotheses

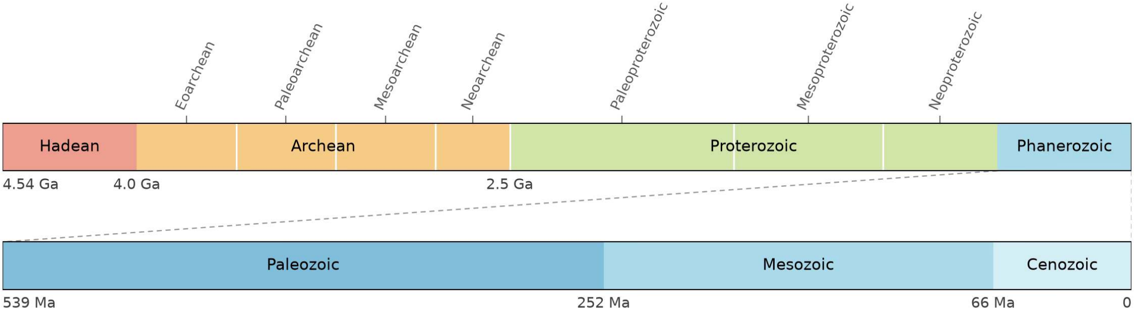
Working definition (Lederberg 1965): a self-replicating, metabolising, evolving chemical system.

Four origin-of-life hypotheses:

- **RNA world**: catalytic ribozymes carry the genetic code before proteins
- **Alkaline hydrothermal vents**: geochemical proton gradients drive early metabolism
- **Warm surface ponds**: wet-dry cycles polymerise organic precursors
- **Panspermia**: transfer of viable organisms between bodies

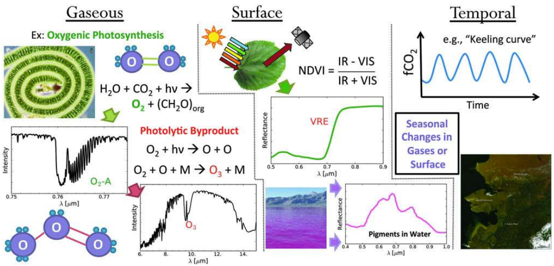


Working definition of life: self-replicating, metabolising, evolving; water and carbon as defaults, other solvents and chemistries not excluded. Course figure.



Geologic eons and eras of Earth across 4.54 Gyr. The Precambrian occupies almost 90% of the timeline, with life emerging early in the Archean before Phanerozoic diversification. Credit: Course-original figure; Gradstein et al. (2020).

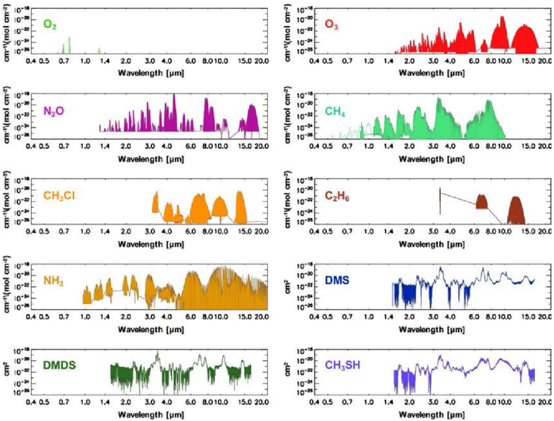
Three Classes of Biosignature



- **Gaseous**: atmospheric O_2, O_3, CH_4, N_2O, and organosulfur species.
- **Surface**: vegetation red edge with sharp reflectance jump near 700 nm.
- **Temporal**: seasonal cycles like Earth's CO_2 Keeling oscillation.
- A credible detection needs several classes together.

Source: Schwieterman et al. (2018)

Biosignature Gases in the Mid-IR



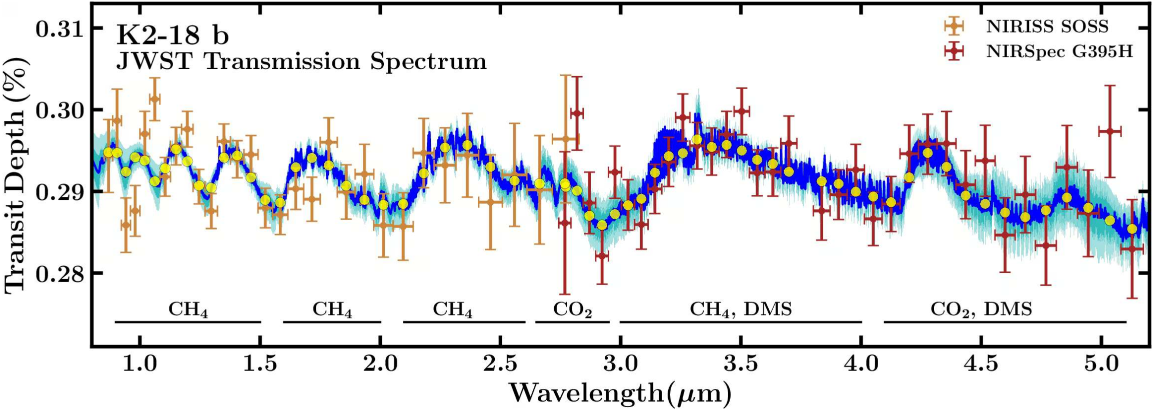
- Ten candidate biosignature gases display characteristic mid-IR bands.
- Key features: $\mathrm{O_3}$ at 9.6 $\mu\mathrm{m}$, $\mathrm{CH_4}$ at 3.3 and 7.7 $\mu\mathrm{m}$.
- $\mathrm{N_2O}$ features and DMS / DMDS window in 3–15 μm.
- These bands set the wavelength needs of LIFE (mid-IR interferometer) and HWO (Habitable Worlds Observatory).

Source: Schwieterman et al. (2018)

Disequilibrium is the Diagnostic

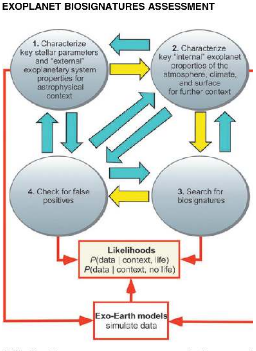
Chemically incompatible gases coexisting in an atmosphere provide the most convincing remote signature of active biological production.

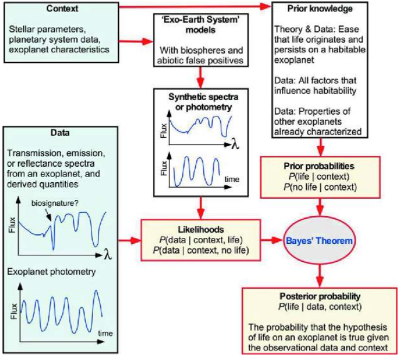
- Individual gases in isolation almost always possess plausible abiotic origins
- Earth pair: $\mathrm{O_2}$ (21%) and $\mathrm{CH_4}$ (1.8 ppm) react on decade timescales
- Simultaneous persistence requires massive, continuous biological resupply fluxes



JWST transmission spectrum of K2-18 b combining NIRISS and NIRSpec observations. Absorption bands reveal atmospheric $\mathrm{CH_4}$ and $\mathrm{CO_2}$ alongside a tentative DMS feature near 3.4 $\mu\mathrm{m}$. Credit: Madhusudhan et al. (2023).

Catling 2018: A Bayesian Framework

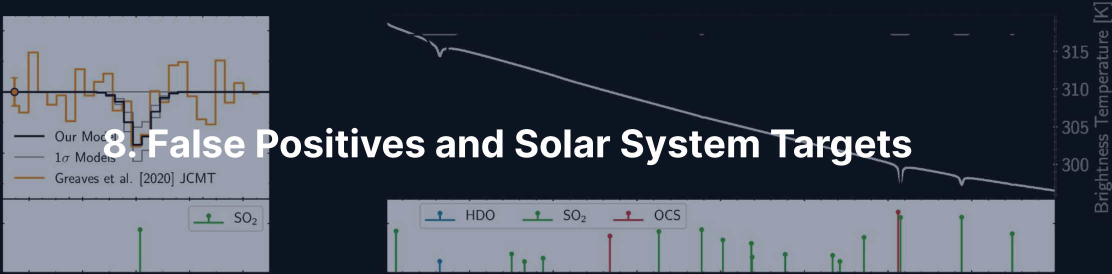




Detection requires a Bayesian chain of inference incorporating stellar context, planetary properties, and abiotic mechanisms, not a single yes/no spectral peak.

Source: Catling et al. (2018)

8. False Positives and Solar System Targets

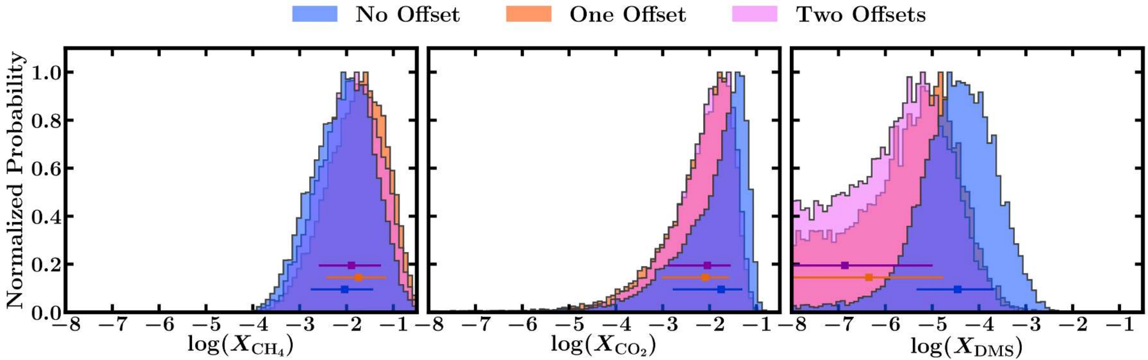


Venus phosphine controversy. Lincowski et al. (2021)

False Positives: Every Gas Has an Abiotic Pathway

Biosignature verification is fundamentally an inverse problem because abiotic mechanisms can produce every individual candidate gas.

- Abiotic $\mathrm{O_2}$: $\mathrm{H_2O}$ photolysis followed by hydrogen escape, or $\mathrm{CO_2}$ photolysis
- Abiotic $\mathrm{CH_4}$: hydrothermal serpentinisation or volcanic outgassing at low oxygen fugacity
- Abiotic $\mathrm{N_2O}$ from lightning and photochemistry; DMS has no significant abiotic Earth source, but one could exist elsewhere
- Confirmation requires ruling out all plausible geochemical and photochemical false positives

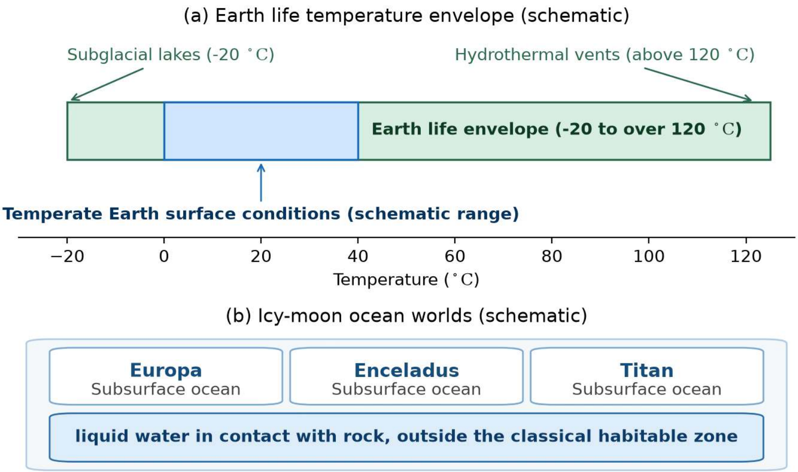


Retrieved mixing ratio posteriors for $\mathrm{CH_4}$, $\mathrm{CO_2}$, and DMS in K2-18 b under zero, one, and two floating instrument offsets. While $\mathrm{CH_4}$ and $\mathrm{CO_2}$ hold, DMS significance collapses from 2.4σ to < 1σ. Credit: Madhusudhan et al. (2023).

Solar System Targets for Life Detection

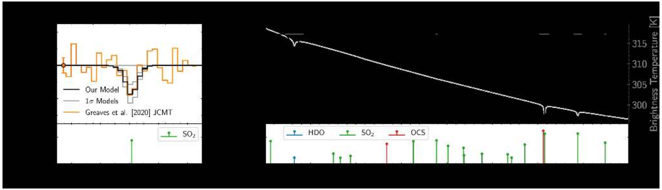
In situ solar system exploration targets five prime environments with liquid water reservoirs or rich organic chemical syntheses.

- Mars: ancient habitable lacustrine environments; Mars Sample Return path
- Europa and Enceladus: confirmed subsurface oceans; Enceladus's plume carries H_2, organics and phosphates from hydrothermal activity, hypothesised for Europa
- Titan and Venus: complex hydrocarbon hydrology and contested cloud chemistry



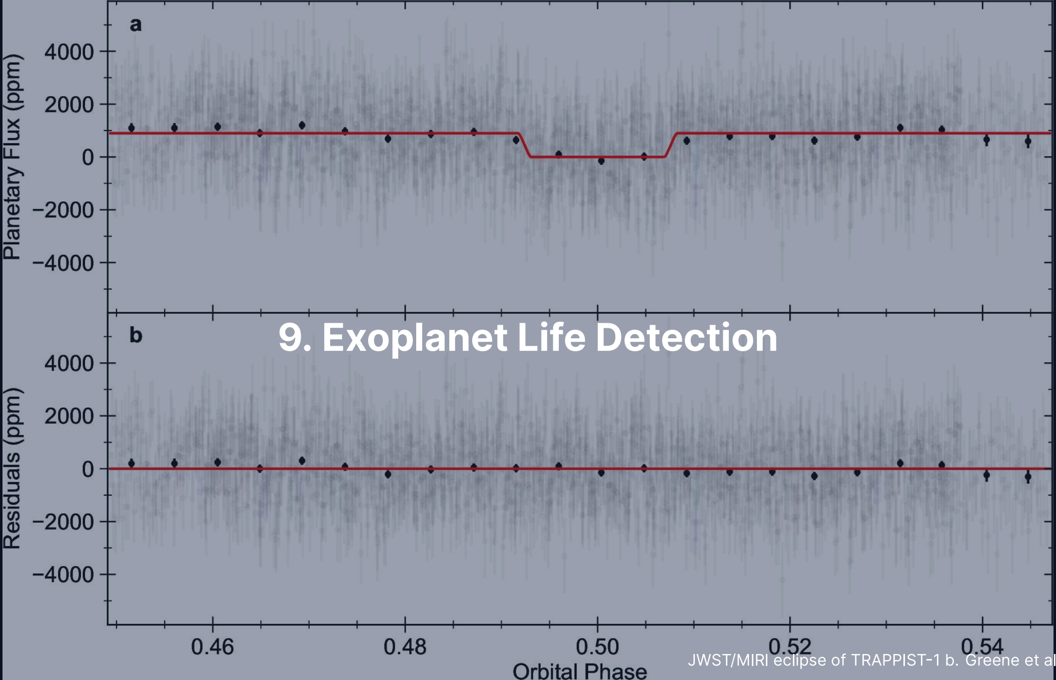
Extremophiles widen the definition of habitable: Earth life spans –20 to over 120 °C, and the icy moons hold liquid water in contact with rock outside the classical habitable zone. Course figure.

Venus Phosphine: A Cautionary Case Study

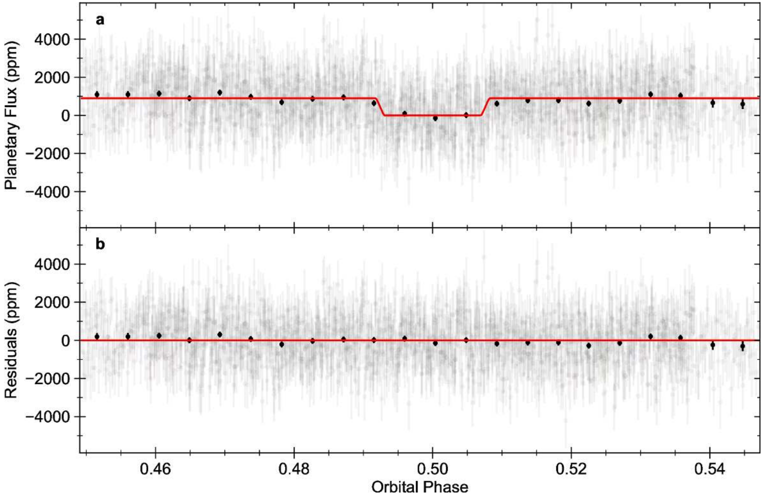


- Greaves et al. (2021): PH_3 in the Venus clouds, a biosignature?
- Lincowski et al. (2021): the line is fit by mesospheric SO_2
- K2-18 b, same lesson: CH_4 (5σ) and CO_2 (3σ) stand; DMS fell from 2.4σ to ~ 1σ
- DAVINCI tests this in situ, EnVision and VERITAS from orbit

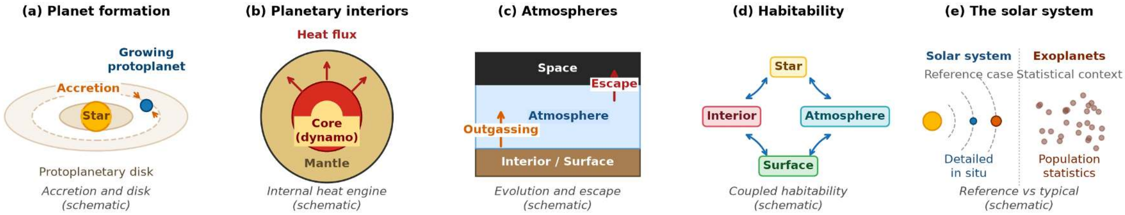
9. Exoplanet Life Detection



JWST/MIRI eclipse of TRAPPIST-1 b. Greene et al. (2023)



JWST MIRI 15 $\mu\mathrm{m}$ secondary eclipse of TRAPPIST-1 b measures a dayside brightness temperature of 503 K, matching the 508 K bare-rock prediction and ruling out a thick atmosphere. Credit: Greene et al. (2023).



The five biggest lessons: formation, interiors as heat engines, dynamic atmospheres, habitability as a coupled property, the solar system as reference case. Course figure.

	Open question	What is at stake, and what will decide it
1	Is the origin of life a rare accident or a generic chemical inevitability?	The largest uncertainty in quantitative habitability estimates
2	What carves the radius valley between super-Earths and sub-Neptunes?	Photoevaporation, core-powered mass loss, or distinct formation channels
3	When did Jupiter form?	Architect of the inner solar system: the NC-CC dichotomy and Earth-like system rarity
4	Was Mars ever inhabited?	Mars Sample Return, the most direct test; it recalibrates the prior on f_l
5	Is the solar system rare or typical?	PLATO, Gaia DR4/DR5 and long-baseline RV give a probabilistic answer

The five biggest open questions, each with what is at stake or the test that will decide it. Course figure.

Summary

1. Planet formation and differentiation are universal physical processes
2. Interiors are heat engines: tectonics, dynamos and outgassing over Gyr
3. Atmospheres evolve through escape, outgassing and photochemistry
4. Habitability is a coupled trajectory, not a static orbital condition
5. The solar system is one detailed reference in a diverse exoplanet population

Next: The final exam.

Thank You

Planetary Systems is complete.

Lecture notes: `formingworlds.github.io/IntroductionToPlanetaryScience`